

Towards Understanding Human-AI Reliance Patterns Through Explanation Styles

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ABSTRACT

As AI is increasingly embedded in human decision-making pipelines, it is necessary to explore the influence of explainable AI (XAI) on both appropriate and inappropriate patterns of human-AI reliance. While various models have been proposed, the impact of explanation style on reliance in the context of these models is under-explored. In particular, there is a lack of research surrounding how different styles of explanations, such as those presented by feature-based and example-based XAI methods, influence how humans interact with and engage with AI-enabled systems through the lens of specific patterns of reliance. We summarise the current literature in this space and conclude by proposing a study and analysis plan that can be used to compare the individual and combined influences of feature-based and example-based explanation styles.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Explainable AI; Human-AI Collaboration; Human-AI Reliance

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1 INTRODUCTION

Increasingly, artificial intelligence (AI) is being introduced into human decision-making pipelines in an attempt to support more accurate performance [11]. However, the increasing complexity of AI models prevents sufficient human understanding of the decision processes underlying AI outputs to support their acceptance at face value [19]. In response, explainable AI (XAI) has been proposed as a method of increasing the interpretability of AI models by providing explanations of their behaviour [19]. On the contrary, much of the

prior literature has found that XAI increases over-reliance on AI [2, 14, 17, 31], leading to poorer decision outcomes overall.

More recently, evidence has suggested that XAI can support appropriate human reliance on AI, provided the explanations are properly engaged with to verify AI decisions [38]. Furthermore, some styles of XAI, whether they be feature-based explanations highlighting important features for the model's decision or example-based explanations displaying examples of similar model inputs and outputs, may influence this decision to engage with explanations and the process of verifying the outputs of the AI system generating them [11]. This research, however, has focused predominantly on over-reliance, disregarding alternative detrimental and beneficial patterns of human-AI reliance and the potential influences of explanation style across this spectrum.

More comprehensive models and metrics of human-AI reliance have since been proposed [4, 35], differentiating between different patterns of reliance by comparing a person's decision before and after being exposed to AI support, however, these metrics have not yet been employed to directly compare the influences of feature-based and example-based explanation styles on human-AI reliance. Furthermore, prior research that has compared feature-based and example-based XAI methods has not considered the effects of combining these explanation styles, thus, an understanding of how they may interact in the context of human-AI reliance has not yet been explored. Given these gaps in the current literature, we propose the following research question to guide our work: Do feature-based and example-based explanations have different influences on beneficial and detrimental patterns of human-AI reliance, and what are the effects of combining these explanation styles?

To address this research question, we propose a study protocol and analysis plan that will be used to examine differences between the beneficial and detrimental reliance patterns demonstrated by participants exposed to feature-based, example-based, and combined feature- and example-based explanations in the context of human-AI collaboration.

2 BACKGROUND AND RELATED WORK

Integrating AI into existing human decision-making pipelines, as opposed to allowing AI systems to operate autonomously, allows AI to make recommendations in high-risk scenarios while retaining human accountability over the outcomes of decisions [1]. Ideally, decisions made through this process of human-AI collaboration should be more accurate than those of either the human or AI alone [1, 11], typically referred to as complementary performance [38]. However, achieving a combined accuracy greater than that of an AI system requires its outputs to be verified by the human

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Table 1: Aligning terminology used in prior research when discussing reliance in human-AI collaboration scenarios. The leftmost three columns state whether the initial human decision, AI decision and final decision are correct (1) or incorrect (0) to produce each reliance pattern, respectively.

Human Decision	AI Decision	Final Decision	Traditional View [38]	Appropriateness View [35]	Dominance View [4]
0	0	0	Over-Reliance		Detrimental Reliance
0	0	1	Appropriate Reliance		Beneficial Under-Reliance
0	1	0	Under-Reliance	Incorrect Self-Reliance	Detrimental Self-Reliance
0	1	1	Appropriate Reliance	Correct AI Reliance	Beneficial Over-reliance
1	0	0	Over-Reliance	Incorrect AI Reliance	Detrimental Over-Reliance
1	0	1	Appropriate Reliance	Correct Self-Reliance	Beneficial Self-Reliance
1	1	0	Under-Reliance		Detrimental Under-Reliance
1	1	1	Appropriate Reliance		Beneficial Reliance

decision-maker to determine whether or not they can be relied upon [31].

2.1 Explainable AI

In response to this need for verification, it has been suggested that improving the interpretability of AI systems through explanations, known as explainable AI (XAI), may improve human-AI collaboration outcomes [11]. XAI methods can be differentiated across various dimensions, resulting in an abundance of tools suitable for a broad range of contexts within and beyond ubiquitous, pervasive, and wearable computing. While exposed model internals can constitute a complete, faithful explanation for simple, glass-box models [34], explanations of complex, black-box models must instead rely on post-hoc approximations [23]. The following discussion will focus on post-hoc explanations due to their broader applicability.

Post-hoc XAI methods differ in the amount of information included in their approximations, from model-transparent methods that have full access to the underlying model [8] to model-agnostic methods that can only access inputs, outputs, and ground-truth labels [18]. Post-hoc explanations can also be global or local in scope. Global explanations provide an overall understanding of the model that is often a more accurate summary of its behaviour [27], while local explanations only extend to individual model outputs and are thus more likely to be inaccurate or inconsistent [19].

XAI methods can additionally be differentiated based on their style of explanation. Feature-based and example-based methods are two common explanation styles seen in XAI. Feature-based explanations calculate the importance, or contribution, of features present in a given input towards the final output of a model, with popular methods including LIME (Local Interpretable Model-Agnostic Explanations) [33] and SHAP (SHapley Additive exPlanations) [25]. Example-based XAI methods provide explanations through prototypical [10], similar [19], or counterfactual examples [29] that aim to allow humans to infer a model's decision boundaries.

Despite the scope of XAI methods proposed in the literature, prior research has repeatedly found that XAI does not reduce [14, 17, 40, 42], or even increases [1, 40], the likelihood that humans will accept incorrect model predictions, typically referred to as over-reliance [11, 38]. In this traditional view of human-AI reliance, summarised in Table 1, only the interaction between a

correct or incorrect AI decision and the final human decision is considered [38]. Thus, over-reliance, and its less frequently observed counterpart under-reliance [38], can be seen under this definition as simply a failure to verify AI decisions appropriately.

Prior work [3, 13, 15, 16, 22] has suggested that this lack of appropriate reliance is due to the mere presence of explanations invoking cognitive biases and heuristics that inevitably lead to blind trust in, and deference to, AI systems. However, more recently it has been proposed that over-reliance is instead a choice arising from weighing the costs and benefits of using an explanation to verify AI [38]. This work by Vasconcelos et al. [38] found evidence to suggest that explanations that are easier for humans to understand, reducing the cost of engaging with them, reduce over-reliance. Thus, determining which styles of explanation are less costly for humans to engage with may assist with developing XAI that is more likely to be used to verify AI decisions and increase appropriate reliance.

2.2 Comparing Explanation Styles

Feature-based explanations have failed to support complementary performance in human-AI collaboration tasks across much of the literature [2, 11, 14, 17, 31], which the traditional view of reliance would suggest is due to a lack of engagement with this style of explanation for verifying AI decisions. Chen et al. [11] investigated this theory using a think-aloud task to compare how participants engaged with feature-based and example-based explanations. They found that the feature-based explanations disrupted participants' reasoning, often emphasising features that led to overthinking or reconsidering their initial correct intuitions. In contrast, the example-based explanations promoted deeper reasoning in which participants utilised similarities between examples to identify features that likely contributed to the model's decision. Additionally, the participants used any differences between examples to identify cases where the prediction made by the AI was unreliable.

These findings suggest the presence of additional factors beyond simply the cost of engaging with an explanation that affect the likelihood of it being used to verify AI outputs. The nature of the think-aloud task used by Chen et al. [11] ensured that participants engaged with the explanations, yet over-reliance was still observed in the feature-based cases. Thus, they proposed that feature-based and example-based explanations may promote different styles of

reasoning with varying efficacy for verifying AI decisions and calibrating reliance appropriately. Measuring these patterns of reliance requires a more complex model than the one offered by the traditional view as, without a measure of human performance on a task both before and after being exposed to AI support, it is impossible to separate reliance on AI from an outcome that may have been observed in its absence [4, 35].

2.3 Measuring Human-AI Reliance

The appropriateness [35] and dominance [4] views of reliance, summarised in Table 1, address the need for a comparison between initial and final human decisions when measuring reliance on AI. Both views define metrics for calculating different types of reliance that can be used to compare AI support interventions, however, we cover only the dominance view in this paper as it considers all of the eight possible reliance patterns differentiated in Table 1.

The dominance view of human-AI reliance, proposed by Cabitza et al. [4], sees reliance in human-AI collaboration as a combination of influences of AI over human decision-making, which can be positive or negative. They define three odds ratios to quantify these influences based on the case-wise frequencies of the eight reliance patterns from Table 1: Technology Impact, Detrimental Algorithmic Aversion, and Automation Bias. Technology Impact is the overall utility of the AI intervention, with values greater than one indicating an overall positive impact. Detrimental Algorithmic Aversion quantifies the positive dominance of the intervention, where values less than one reflect a tendency for humans to accept correct AI decisions. Finally, Automation Bias quantifies the negative dominance of the intervention, where values greater than one reflect a tendency for humans to accept incorrect AI decisions. Both frequentist and causal versions of the Detrimental Algorithmic Aversion and Automation Bias odds ratios have been defined, allowing the direct causal influences of AI and XAI support on reliance to be quantified [4].

These dominance metrics were subsequently used to interpret the findings of prior human-AI collaboration studies [5, 6, 32] where the AI support provided either no explanation or feature-based explanations. These analyses identified a higher degree of negative dominance for AI support with explanations compared to no explanations, aligning with previous research in which feature-based XAI has typically increased traditional over-reliance [2, 11, 14, 17, 31]. Conversely, little evidence was found to support a difference in positive dominance between the presence and absence of XAI. It remains unclear whether the patterns of reliance on feature-based explanations measured in these studies extend to other styles of explanations.

3 METHOD

To address our research question we plan to conduct a two-factor mixed randomised experiment with four between-subjects explanation style conditions (no explanations, feature-based explanations, example-based explanations, and combined feature- and example-based explanations) and two within-subjects task conditions (image and textual). We will use SHAP [25] to generate the feature-based explanations and ExMatchina [19] to generate the example-based explanations, which will have their ground-truth labels removed

to avoid confounding effects from attributes outside of the examples themselves. The combined explanations will be generated by applying SHAP to the two nearest examples determined by ExMatchina. We have observed that the SHAP feature attributions for these examples and the input case they are generated for are visually similar using this combination method, which we take as an indication of approximately equivalent information between our proposed explanation-style conditions.

Following prior research [1, 30, 38] participants will be recruited via online crowd-sourcing, specifically, Amazon Mechanical Turk (MTurk). To ensure a diverse participant pool and increase the generalisability of our research [9, 30], we will ask participants their age, gender, level of education, and country of residence prior to beginning the study.

3.1 Protocol

Our proposed study protocol consists of two parts, outlined in Figure 1. Part 1 asks participants to perform 12 trials of two human-AI collaboration tasks (six of each task) both with and without AI support, allowing us to measure the effect of introducing this support on human performance [4, 35]. The order of presentation of these tasks will be counterbalanced, with the order of case presentation within each task randomised across participants.

The first task is a fine-grained bird image classification task using images of four wren species drawn from the Caltech-UCSD Birds (CUB) dataset [39], which has been used in prior human-AI collaboration research [10, 21, 24, 28, 37, 41], setting a benchmark for our work. We have chosen bird image classification over more complex tasks, such as medical image classification [7], due to its greater approachability for lay users [28]. The second task requires participants to determine the truthfulness of statements generated by the large language model InstructGPT, classifying them as either correct or incorrect [12]. These two tasks combined provide a good representation of the types of AI outputs (visual and textual) that may be encountered by the users of ubiquitous computing systems. The cases presented to participants will be controlled such that the accuracy of the AI in the study is 66% (four out of six correct per task), which is approximately equivalent to human baseline accuracy on these tasks (as measured in a pilot study) to ensure that neither over- nor under-relying on the AI is inherently more beneficial [1, 38]. To calculate the causal metrics proposed by Cabitza et al. [4] we will stratify the cases used in the study based on image quality and sentence complexity.

Since individual differences in trust are a confounding factor when measuring reliance [38], we will adopt the same controls for mitigating trust used by Vasconcelos et al. [38]. Namely, the AI presented to participants will not be anthropomorphised, its outputs will be presented as suggestions that do not need to be accepted, and its level of accuracy will not be made known to participants beyond stating that it is equivalent to average human accuracy. Additionally, participants will not be informed of their own accuracy on either task until the completion of the study to prevent them from using this information to update their perception of the AI's accuracy between trials [38].

In Part 2 of our study, participants will be required to predict the AI's suggestions on four new cases (two image and two textual)

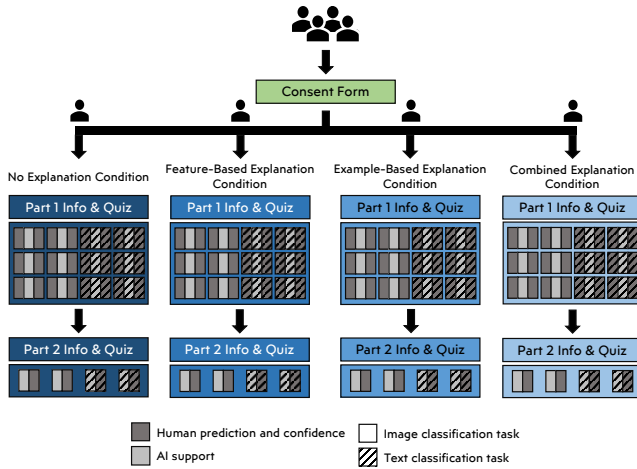


Figure 1: Proposed mixed study protocol. Participants will be randomly allocated to one of four explanation-style groups after reading and agreeing to the consent form and will be assigned to the same explanation condition for both Part 1 and Part 2. Before beginning each Part, participants will be provided information about the image and text classification tasks followed by a short quiz to confirm their understanding.

based on their interactions with the AI in Part 1 and its explanations for these new cases, depending on the condition the participant was assigned to in Part 1. This part of the study aims to measure how effectively different explanation styles increase interpretability through human understanding of an AI’s behaviour [26] and their ability to predict its future behaviour [20].

3.2 Analysis

Our analysis of Part 1 will primarily involve calculating the odds ratio metrics defined by Cabitza et al. [4], allowing us to compare the influences of each explanation-style condition on beneficial versus detrimental reliance (Technology Impact), positive dominance (Detrimental Algorithmic Aversion), and negative dominance (Automation Bias). We will supplement this analysis by calculating the metrics corresponding to the appropriateness view of reliance proposed by Schemmer et al. [35], evaluating any differences between the inferences that can be made from each group of metrics. In our analysis of Part 2, we will compare participants’ accuracy across the four conditions, noting any differences between when the AI’s suggestion matches the true class label and when it does not. We will use this analysis to contextualise our findings regarding human-AI reliance from Part 1.

In both Parts 1 and 2, participants will be asked to rate their confidence in each answer on a validated 5-point Likert scale [36]. This will allow us to observe cases where AI support has a smaller impact on human decision-making, increasing or decreasing a participant’s confidence in their answer without resulting in a change in decision. This additional measurement will be particularly useful for differentiating between correct initial decisions with high versus low confidence, as this has implications regarding the degree of

influence of the AI over a final decision that is also correct, which is something that cannot be measured by the metrics we will be using as part of our main analysis [4, 35].

While there is insufficient consensus amongst prior literature to allow us to propose formal hypotheses for the findings of our proposed analyses, we expect to see differences between the no explanation and explanation conditions. In particular, we expect to observe correlations between the dominance of combined explanations and that of feature-based and example-based explanations individually, where the strengths of these correlations are dependent upon the relative influence of each explanation style on human-AI reliance.

4 CONCLUSIONS AND FUTURE WORK

In this paper, we summarised the current state of the literature surrounding reliance in human-AI collaboration, identifying a lack of research regarding the role of explanation style on distinct patterns of reliance. We began to address this gap by proposing a study protocol and analysis plan that we will use in future work to compare the influences of feature-based, example-based, and combined feature- and example-based explanations on human-AI reliance, compared to a baseline of AI predictions without XAI support, in tasks involving both image and textual cases. We anticipate that this work will contribute towards recognising and improving the qualities of XAI that promote more positive human-AI collaboration outcomes in tasks representative of a broad range of ubiquitous computing scenarios.

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